Circuit Analytics Results

case1_bacillus

FakeTorino

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	6	3	26
With Start End	L2	6	3	25
With Start End	L3	6	3	25
Without Start End	L1	12	3	26
Without Start End	L2	12	3	25
Without Start End	L3	12	3	25
Custom Ansatz	L1	3	3	34
Custom Ansatz	L2	3	3	33
Custom Ansatz	L3	3	3	33

FakeFez

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	6	3	26
With Start End	L2	6	3	25
With Start End	L3	6	3	25
Without Start End	L1	12	3	26
Without Start End	L2	12	3	25

Without Start End	L3	12	3	25
Custom Ansatz	L1	3	3	34
Custom Ansatz	L2	3	3	33
Custom Ansatz	L3	3	3	33

case2_genome

FakeTorino

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	27	3	34
With Start End	L2	27	3	33
With Start End	L3	27	3	33
Without Start End	L1	33	3	34
Without Start End	L2	33	3	33
Without Start End	L3	33	3	33
Custom Ansatz	L1	35	35	162
Custom Ansatz	L2	35	35	161
Custom Ansatz	L3	35	35	161

FakeFez

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	27	3	34
With Start End	L2	27	3	33
With Start End	L3	27	3	33
Without Start End	L1	33	3	34
Without Start End	L2	33	3	33
Without Start End	L3	33	3	33
Custom Ansatz	L1	35	35	162

Custom Ansatz	L2	35	35	161
Custom Ansatz	L3	35	35	161

case3_bacteriodes

FakeTorino

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	10	2	30
With Start End	L2	10	2	29
With Start End	L3	10	2	29
Without Start End	L1	14	2	30
Without Start End	L2	14	2	29
Without Start End	L3	14	2	29
Custom Ansatz	L1	14	14	78
Custom Ansatz	L2	14	14	77
Custom Ansatz	L3	14	14	77

FakeFez

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	10	2	30
With Start End	L2	10	2	29
With Start End	L3	10	2	29
Without Start End	L1	14	2	30
Without Start End	L2	14	2	29
Without Start End	L3	14	2	29
Custom Ansatz	L1	14	14	78

Custom Ansatz	L2	14	14	77
Custom Ansatz	L3	14	14	77

case4_avian_flu

FakeTorino

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	64	4	38
With Start End	L2	64	4	37
With Start End	L3	64	4	37
Without Start End	L1	72	4	38
Without Start End	L2	72	4	37
Without Start End	L3	72	4	37
Custom Ansatz	L1	79	79	338
Custom Ansatz	L2	79	79	337
Custom Ansatz	L3	79	79	337

FakeFez

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	64	4	38
With Start End	L2	64	4	37
With Start End	L3	64	4	37
Without Start End	L1	72	4	38
Without Start End	L2	72	4	37
Without Start End	L3	72	4	37
Custom Ansatz	L1	79	79	338

Custom Ansatz	L2	79	79	337
Custom Ansatz	L3	79	79	337

case5_denv2

FakeTorino

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	8	2	30
With Start End	L2	8	2	29
With Start End	L3	8	2	29
Without Start End	L1	12	2	30
Without Start End	L2	12	2	29
Without Start End	L3	12	2	29
Custom Ansatz	L1	11	11	66
Custom Ansatz	L2	11	11	65
Custom Ansatz	L3	11	11	65

FakeFez

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	8	2	30
With Start End	L2	8	2	29
With Start End	L3	8	2	29
Without Start End	L1	12	2	30
Without Start End	L2	12	2	29
Without Start End	L3	12	2	29
Custom Ansatz	L1	11	11	66

Custom Ansatz	L2	11	11	65
Custom Ansatz	L3	11	11	65

case6_monkeypox

FakeTorino

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	21	3	34
With Start End	L2	21	3	33
With Start End	L3	21	3	33
Without Start End	L1	27	3	34
Without Start End	L2	27	3	33
Without Start End	L3	27	3	33
Custom Ansatz	L1	27	27	130
Custom Ansatz	L2	27	27	129
Custom Ansatz	L3	27	27	129

FakeFez

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	21	3	34
With Start End	L2	21	3	33
With Start End	L3	21	3	33
Without Start End	L1	27	3	34
Without Start End	L2	27	3	33
Without Start End	L3	27	3	33
Custom Ansatz	L1	27	27	130

Custom Ansatz	L2	27	27	129
Custom Ansatz	L3	27	27	129

case7_african_swine_21

FakeTorino

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	76	4	38
With Start End	L2	76	4	37
With Start End	L3	76	4	37
Without Start End	L1	84	4	38
Without Start End	L2	84	4	37
Without Start End	L3	84	4	37
Custom Ansatz	L1	94	94	398
Custom Ansatz	L2	94	94	397
Custom Ansatz	L3	94	94	397

FakeFez

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	76	4	38
With Start End	L2	76	4	37
With Start End	L3	76	4	37
Without Start End	L1	84	4	38
Without Start End	L2	84	4	37
Without Start End	L3	84	4	37
Custom Ansatz	L1	94	94	398

Custom Ansatz	L2	94	94	397
Custom Ansatz	L3	94	94	397

case8_african_swine_24

FakeTorino

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	88	4	38
With Start End	L2	88	4	37
With Start End	L3	88	4	37
Without Start End	L1	201	28	96
Without Start End	L2	156	21	89
Without Start End	L3	162	24	89
Custom Ansatz	L1	235	232	720
Custom Ansatz	L2	262	241	799
Custom Ansatz	L3	210	210	709

FakeFez

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	88	4	38
With Start End	L2	88	4	37
With Start End	L3	88	4	37
Without Start End	L1	96	4	38
Without Start End	L2	96	4	37
Without Start End	L3	96	4	37
Custom Ansatz	L1	109	109	458

Custom Ansatz	L2	109	109	457
Custom Ansatz	L3	109	109	457

case9_hantavirus_31

FakeTorino: Exceeds backend qubit count: requires 155, backend has 133

FakeFez

Circuit Type	Opt Level	2Q Gates	2Q Gate Depth	Total Depth
With Start End	L1	116	4	38
With Start End	L2	116	4	37
With Start End	L3	116	4	37
Without Start End	L1	247	31	108
Without Start End	L2	235	27	110
Without Start End	L3	197	29	123
Custom Ansatz	L1	570	564	1467
Custom Ansatz	L2	503	479	1459
Custom Ansatz	L3	421	404	1234